

Participation Activity 9

Solution

Last Name: _____

First Name: _____

Std #: _____

List the micro-instructions which are executed during each of the following five stages when the machine instruction `sw $s5, 8($sp)` is executed.

instruction:
instruction format:
machine code:

`sw $s5, 8($sp)` **Meaning:** $M[8 + R[\$sp]] \leftarrow R[\$s5]$

Instruction composed of 1 word (32 bits)

opcode	rs	rt	Immediate or address
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101011	10101	11101	0000 0000 0000 1000
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Instruction register IR

1. Fetch (F) $\longrightarrow$
 - 1. $IR \leftarrow M[PC]$ $\rightarrow$ fetching 1 word worth of instruction
 - 2. $PC \leftarrow PC + 4 \text{ bytes}$ $\rightarrow$ next instruction is 1 word (4 bytes) further in memory
2. Decode (D) $\longrightarrow$
 - 1. $valA \leftarrow R[10101]$ i.e., $R[21] \rightarrow \$s5$
 - 2. $valB \leftarrow R[11101]$ i.e., $R[29] \rightarrow \$sp$
 - 3. $valI \leftarrow 8$ (0000 0000 0000 1000)
3. Execute (E) $\longrightarrow$ $valE \leftarrow valB + valI$
4. Memory (M) $\longrightarrow$ $M[valE] \leftarrow valA$
5. Write back (W) $\longrightarrow$ **Nothing is done in this stage!**

OR
 $valA \leftarrow R[29]$
 $valB \leftarrow R[21]$